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EARLYSIGN: EARLY AUTISM RISK SCREENING USING BEHAVIOURAL AND BIOLOGICAL BIOMARKERS

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BONAFIDE CERTIFICATE

Certified that this project report "**EARLYSIGN: EARLY AUTISM RISK SCREENING USING BEHAVIOURAL AND BIOLOGICAL BIOMARKERS**" is the bonafide work of **HARNESH G – 24623502**, **KISHORE R – 24623140** and **DHARUN KUMAR R – 2462501**. They carried out this project under the guidance and supervision of the Department of Artificial Intelligence and Machine Learning, School of Engineering, Vels Institute of Science, Technology & Advanced Studies (VISTAS), Chennai.

The report explains the design and the prototype of a screening tool for autism in young children. The tool brings eight kinds of early signs into one place, gives a risk level of Low, Medium or High, and explains how it reached that level.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

Autism is often found late. Early signs can show up between 12 and 24 months of age, but many children are diagnosed only at three or four years. By then some of the best years for early support have passed. Most screening today depends on a parent questionnaire and a short visit to a doctor. No single test is good enough to screen every child.

EarlySign is a browser-based prototype that brings up to eight kinds of early signs into one place. These are a parent questionnaire, eye tracking, EEG, MRI, blood markers, genetic results, speech and motor skills. Each module is optional. The tool turns every measure into a score from 0 to 100%, then combines them in three ways: a simple weighted score, a logistic regression model and a small decision tree. The result is an ASD risk level of Low, Medium or High, with a chart of what drove the score and a list of warning signs in plain words. The tool can also compare two children and raise an alert when a child's score rises sharply between visits.

We trained and tested the two machine learning models on 300 simulated children, keeping 90 aside for testing. On this test set the weighted score reached an AUC of 0.89, and both models reached about 0.85. These numbers come from made-up data, so they show that the method works, not how well it would work in a clinic. The tool is a research prototype. It cannot diagnose autism.

Keywords: Autism Spectrum Disorder, Early Screening, Biomarkers, Eye Tracking, EEG, M-CHAT, Logistic Regression, Decision Tree, Explainable AI, Web Application.

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Abbreviation	Full Form
AI	Artificial Intelligence
ASD	Autism Spectrum Disorder
AUC	Area Under the ROC Curve
CART	Classification and Regression Tree
CSF	Cerebrospinal Fluid
CSV	Comma-Separated Values
EEG	Electroencephalography
HOD	Head of the Department
HTML	HyperText Markup Language
IL-6	Interleukin 6
JSON	JavaScript Object Notation
M-CHAT-R/F	Modified Checklist for Autism in Toddlers, Revised with Follow-up
ML	Machine Learning
MRI	Magnetic Resonance Imaging
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SVG	Scalable Vector Graphics
TNF-alpha	Tumour Necrosis Factor alpha
UI	User Interface
VISTAS	Vels Institute of Science, Technology and Advanced Studies

CHAPTER 1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Autism spectrum disorder (ASD) is a lifelong condition that changes how a child talks, plays and connects with other people. About 1 in 100 children in the world has autism [12], and in the United States the figure is 1 in 36 children aged eight [10]. Signs can appear before the second birthday. Even so, many families wait years for a diagnosis.

Finding autism early matters because the young brain changes quickly. A child who gets support at two years old has more time to build skills than a child who starts at five. The difficulty is that early signs are small and easy to miss in a short check-up.

EarlySign is a screening prototype that we built to help with this. A doctor or researcher enters whatever test results are available for a child. The tool turns them into one ASD risk level, explains which results pushed the level up, and suggests the next step. The idea for the project comes from the review by A. Puri on early detection of autism with the help of biomarkers [1]. A biomarker is something we can measure, such as where a child looks, brain activity or a blood value.

The whole tool is a single web page. It runs in any modern browser, needs no installation and sends no data to a server. All the children shipped with the app are simulated, so no real patient data is used anywhere in this project.

1.2 PROBLEM STATEMENT

Screening for autism today has four main problems. First, it happens late. Second, it is subjective, because questionnaires depend on what a parent remembers and notices. Third, no single biomarker is accurate enough on its own. Fourth, the results that do exist are spread out. A child may have an eye-tracking result in one lab, an EEG in another and a questionnaire in a clinic file, and nobody looks at them together.

So our question was simple. Can we put several early signs in one place, work with whatever tests a clinic happens to have, explain the result clearly, and help a doctor decide sooner which toddlers should get a full assessment?

1.3 OBJECTIVES

- To combine up to eight early-sign modules into one ASD risk level (Low, Medium or High) for children aged 12 to 36 months.

- To give a score even when some tests are missing, and to show how much data the score is based on.
- To compare a simple weighted score with two machine learning models: logistic regression and a decision tree.
- To explain every result module by module, in plain language a parent could follow.
- To track the same child over repeat visits and raise an alert when the risk rises sharply.
- To keep all data on the user's own device.

1.4 SCOPE OF THE PROJECT

The prototype covers children from 12 to 36 months old. It has a dashboard, a form to add a child, a step-by-step assessment, a short questionnaire for parents, a screening report, a page to compare two children, a settings page and a guided live demo. It works with eight modules: behaviour, eye tracking, EEG, MRI, blood, genetic, speech and motor.

Some things are outside the scope. The tool does not read raw EEG signals, MRI scans or audio recordings. It takes summary numbers that a lab would already have worked out. The reference ranges and weights are placeholders that we chose, not clinical norms. The models are trained on simulated children because we did not have access to real clinical data. For these reasons EarlySign is a research and teaching prototype, and every result screen says that it is not a diagnosis.

1.5 APPLICATIONS AND EXPECTED OUTCOMES

- Child development clinics that want one view of the screening results they already collect.
- Research groups that study how different biomarkers add up.
- Teaching, to show students how a screening score is built and explained.
- Follow-up of high-risk children, such as younger siblings of a child with autism, over several visits.

We expect the tool to make scattered results easier to read together, to make the reasons behind a score visible, and to show how a simple weighted score compares with machine learning on the same data. Real clinical use would need proper reference values, testing on real children and approval from an ethics committee.

CHAPTER 2. LITERATURE REVIEW

2.1 BACKGROUND

Autism is diagnosed by watching behaviour, using tools such as clinical observation and interviews [9]. There is no blood test or scan that confirms it. Researchers have spent years looking for early measurable signs, because behaviour alone often becomes clear only after the second birthday. These measurable signs are called biomarkers.

Our main reference is the review “Early detection of autism with the help of biomarkers: A narrative review” by A. Puri [1]. It looks at biomarkers that could help find autism early. EarlySign uses the same broad groups of signs and puts them into one working screen.

2.2 EXISTING APPROACHES

Questionnaires. The M-CHAT-R/F is a 20-question checklist that parents fill in for children aged 16 to 30 months, followed by a short interview when needed [2]. It is quick, cheap and widely used. Its weakness is that it depends on what the parent notices, and fixed cut-offs miss some children.

Eye tracking. Jones and Klin followed babies from 2 to 24 months and found that attention to the eyes of other people dropped in infants who were later diagnosed with autism [3]. Pierce and others found that some toddlers with autism prefer moving geometric patterns to videos of people [4]. These are objective signals, but they need special equipment.

Brain imaging and EEG. MRI studies have found extra fluid around the brain (extra-axial CSF) and faster growth of brain surface area in infants who were later diagnosed [5], [6]. EEG is cheaper than MRI, and Bosl and others used EEG features with machine learning to separate infants at high and low risk [7]. Both methods are still mostly used in research labs.

Blood and genetic markers. Raised blood serotonin is one of the oldest findings in autism research [14]. Signs of mitochondrial problems have also been reported [13]. Some rare gene changes, such as those linked to Fragile X syndrome, carry a high chance of autism, and common gene variants add a small amount of risk [15]. Family history matters too. A younger sibling of a child with autism has about a 1 in 5 chance of also having it [11].

Speech and movement. Automatic analysis of a child's voice recordings has been used to tell children with autism apart from other children [16]. Head lag when a baby is pulled up to sit has also been studied as an early sign [17].

Each of these methods looks at one kind of signal. Most of them were tested in research settings with special equipment.

2.3 RESEARCH GAP

We found three gaps. The first is that most tools look at one kind of signal only. The second is that a clinic rarely has every test, so a tool that needs a full set of results is not practical. The third is that machine learning models often give a number without saying why. A doctor is unlikely to act on a score they cannot check.

There is also a smaller gap. Development changes fast in the first three years, so one assessment can mislead. A tool should make it easy to see whether a child's risk is going up or down over several visits.

2.4 PROPOSED CONTRIBUTION

- One screen that brings together eight kinds of early signs.
- A score that works with any subset of modules and says how much data it used.
- Three scoring methods shown side by side, so it is easy to see when they disagree.
- An explanation for every score, with warning signs written in plain words.
- Repeat visits, comparison of two children and an alert when risk rises sharply.

We do not claim that EarlySign is accurate on real children. Our contribution is a working design that shows how these parts can fit together, tested on simulated data.

CHAPTER 3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

EarlySign has four layers. The input layer collects the child's details and test results. The processing layer turns each result into a score. The model layer combines the scores. The output layer shows the risk level, the explanation and the follow-up tools. All data is kept in the browser's own storage. Figure 1 shows the layers.

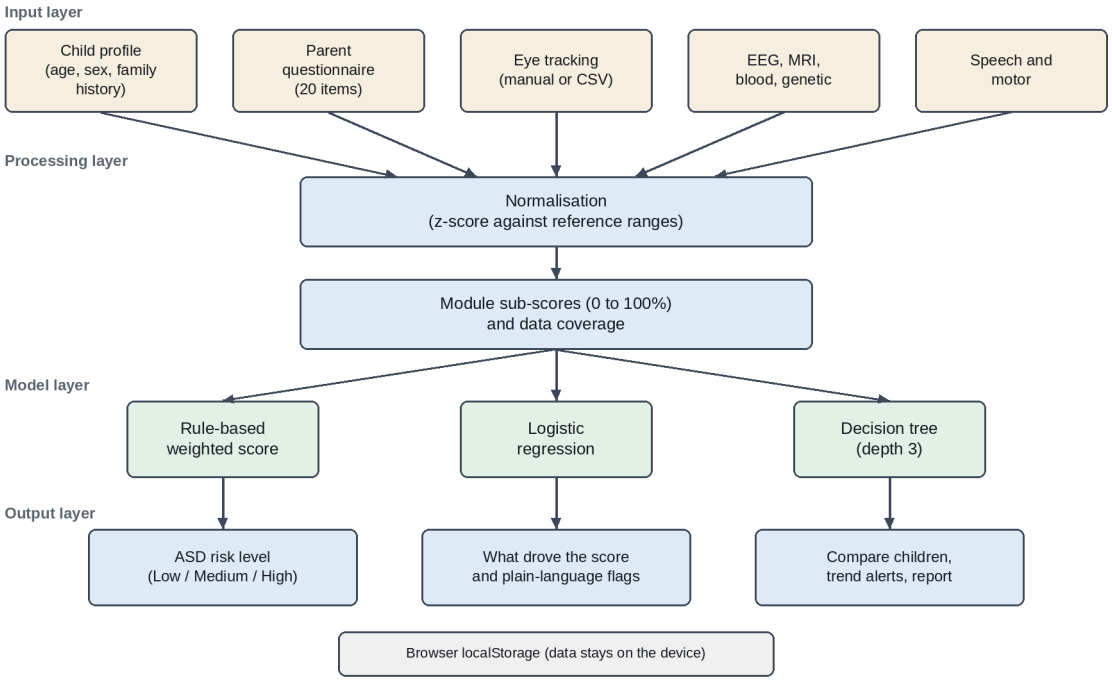


Figure 1 - Overall architecture of EarlySign

Table 1 - Layers of the system

Layer	Main part	Work done
Input	Child profile and 8 modules	Takes age, sex, family history and any test results the clinic has.
Processing	Normalisation	Compares each value with a reference range and turns it into a z-score.
Processing	Module sub-scores	Turns each module into a score from 0 to 100% and records how complete it is.
Model	Three scoring methods	Weighted score, logistic regression and decision tree combine the sub-scores.
Output	Report and follow-up	Shows the ASD risk level, what drove it, next steps, comparisons and trend alerts.
Storage	Browser localStorage	Keeps the children and settings on the device. Nothing is sent to a server.

3.1.1 Overall Data Flow

1. The user creates a child profile with a pseudonymous ID, age, sex, gestation and family history.
2. The user enters results for any of the eight modules. Any step can be skipped.
3. Each measure is compared with its reference range.
4. Each module gets a sub-score from 0 to 100%.
5. The three scoring methods combine the sub-scores into one risk score.
6. The score is turned into an ASD risk level of Low, Medium or High.
7. The tool works out which modules drove the score.
8. The report shows the result, next steps and any trend alert.

Figure 2 shows this flow from the child profile to the report.

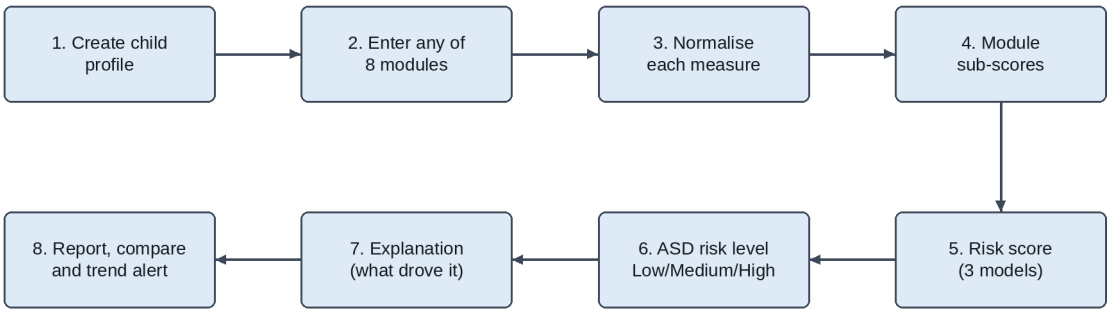


Figure 2 - Data flow from child profile to report

3.2 BIOMARKER MODULES

Table 2 lists the eight modules, what each one measures and its default weight in the weighted score. The weights add up to 100% and can be changed on the settings page.

Table 2 - Biomarker modules

No.	Module	What we measure	Weight
1	Behaviour	20-question parent questionnaire in the style of the M-CHAT	25%
2	Eye tracking	Time spent on faces, social versus geometric scenes, speed of following another person's gaze	18%
3	EEG	Alpha power, theta/alpha ratio, gamma power, long-range coherence	12%
4	MRI	Extra-axial CSF, growth of brain volume, growth of surface area	10%
5	Blood	Oxidative stress, serotonin, lactate/pyruvate ratio, IL-6, TNF-alpha	7%
6	Genetic	Fragile X, CHD8, SHANK3, 16p11.2, PTEN, SCN2A and a polygenic score	10%
7	Speech	Babbling, pitch variability, delay in responding to name	10%
8	Motor	Head lag, postural control, movement variability	8%

For every measure we store a reference mean, a standard deviation and a direction. The direction says whether a higher or a lower value means more risk.

For example, less time looking at faces means more risk, while a longer delay in following gaze means more risk. The reference values in the app are our own placeholders and are marked as such on the screen.

3.3 SOFTWARE ARCHITECTURE

EarlySign is one HTML file of about 1,400 lines. It uses plain HTML, CSS and JavaScript with no outside libraries, so it also works offline. Charts are drawn as SVG. The live demo uses the browser's Web Speech API for the optional voice-over. Table 3 lists the main parts.

Table 3 - Software parts of the system

Software part	Purpose in the system
Router and views	Shows the dashboard, child page, assessment form, report, compare, settings and about pages.
Scoring engine	Turns raw values into z-scores, module sub-scores and the final risk score.
Model code	Trains logistic regression and the decision tree in the browser and works out the test results.
Synthetic data generator	Makes 300 simulated children for training and six demo children for the dashboard.
Chart functions	Draw the bar charts, the risk-over-time line chart and the calibration chart in SVG.
Storage	Saves children and settings in localStorage, with export and import as JSON.
Live demo	A 19-step guided tour with an optional spoken voice-over.

3.4 SCORING DESIGN

Scoring happens in three steps. In the first step every measure is turned into a z-score, which says how far the value is from the reference mean, counted in standard deviations. We multiply by the direction so that a positive z always means “further towards risk”. A measure with z of 1.5 or more is listed as a warning sign in the report.

In the second step each module gets a sub-score. We take the average z of the measures that were filled in and pass it through a logistic curve, which squeezes any number into the range 0 to 100%. The questionnaire works differently. We count the answers that point to risk, scale the count to 20 questions and pass that

through its own curve. For the genetic module, a known high-impact gene change sets the sub-score to at least 85%.

In the third step the sub-scores are combined. The weighted score is the weighted average of the modules that were done, so missing modules are left out instead of counted as zero. A child with a sibling who has autism gets 5 points added, and a child with other family history gets 3 points. The final score is turned into a risk level using the cut-offs in Table 4.

Table 4 - ASD risk levels

Risk level	Score	Suggested next step
Low	Below 35%	Routine check-ups and screening again at the next visit.
Medium	35% to 59%	Review again in three months, add missing modules and talk with the family about support.
High	60% and above	Refer for a full diagnostic assessment and consider early support now.

The report also shows data coverage. This is the share of the total weight that was actually used. A High result based on two modules is less certain than one based on all eight, and the report says so.

3.5 MACHINE LEARNING DESIGN

We added two machine learning models to see whether they would do better than the weighted score. Both take the eight module sub-scores as input. A missing module is set to a neutral 50%.

Logistic regression. This model learns one weight for each module and adds them up. We wrote it ourselves in JavaScript and train it with gradient descent for 2,000 rounds when the page loads. Because the model is a straight sum, we can show exactly how much each module pushed the score up or down compared with an average child. These are the Shapley values of the model [8].

Decision tree. This model asks at most three yes/no questions about the module scores, such as “is the behaviour score above 61%?”. It splits the children using Gini impurity, which is the usual CART method, and needs at least 10 children in every leaf. To explain a result, the report shows the path the child took through the tree and how much the risk changed at each step.

Training data. We had no real patient data, so the app makes 300 simulated children with a fixed random seed. About 30% are labelled ASD. Autism looks different in every child, so each simulated child with an ASD label gets a random severity that controls how far their values move from the reference. A quarter of the modules are left out at random, to copy how real clinics have missing tests. We train on the first 210 children and test on the other 90, which the models never see during training.

CHAPTER 4. IMPLEMENTATION

4.1 DASHBOARD

The dashboard is the first screen. It shows how many children are at each risk level and lists every child with their age, last assessment, number of modules done, score and risk level. The user can search by ID, filter by risk level and tick two children to compare them. An upward arrow appears next to a child whose score rose sharply since the last visit.

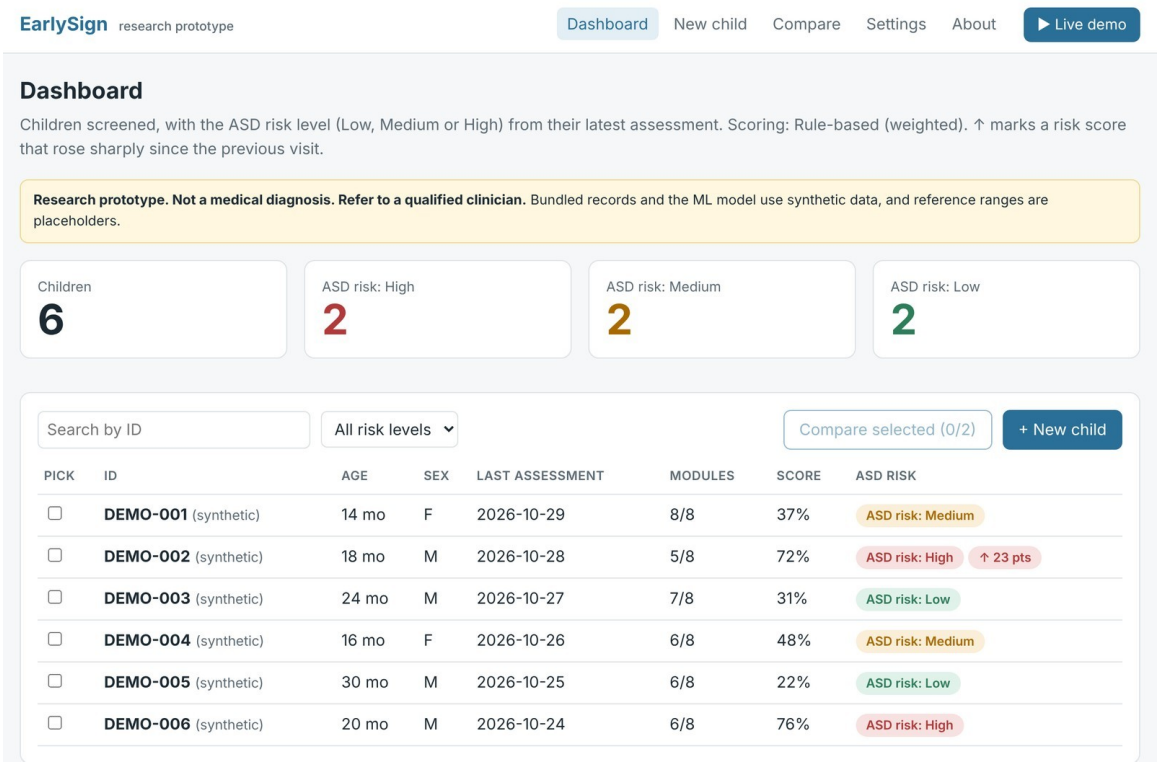


Figure 3 - Dashboard with ASD risk levels and a trend arrow

In Figure 3, DEMO-002 shows an arrow with “23 pts”. This child's score went from 49% to 72% between the last two visits, which is above the alert limit of 15 points. The six demo children are simulated and are marked “synthetic” in the list. The list is drawn by viewDashboard and the arrow comes from trendAlert.

4.2 CHILD PROFILE AND ASSESSMENT FORM

A new child is added with a pseudonymous ID, age in months, sex, gestational age and two family history boxes. The form tells the user not to enter names. After saving, the app opens the assessment form straight away.

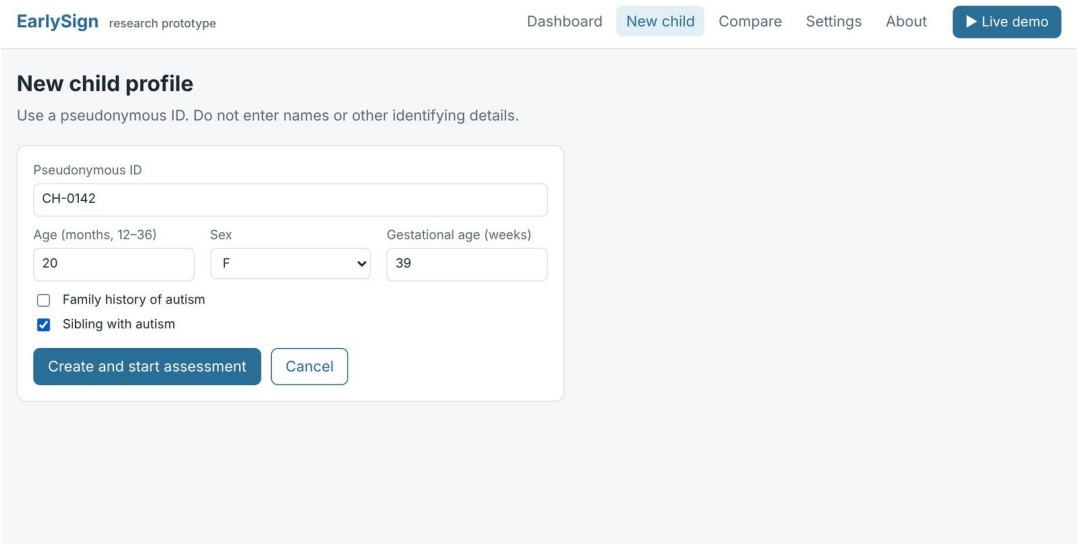


Figure 4 - New child profile

The assessment form has one step per module and a review step at the end. The buttons at the top show which steps are done and let the user jump between them. Figure 5 shows the behaviour step. The questions follow the style of the M-CHAT but are written in our own words. At least 10 answers are needed for a score.

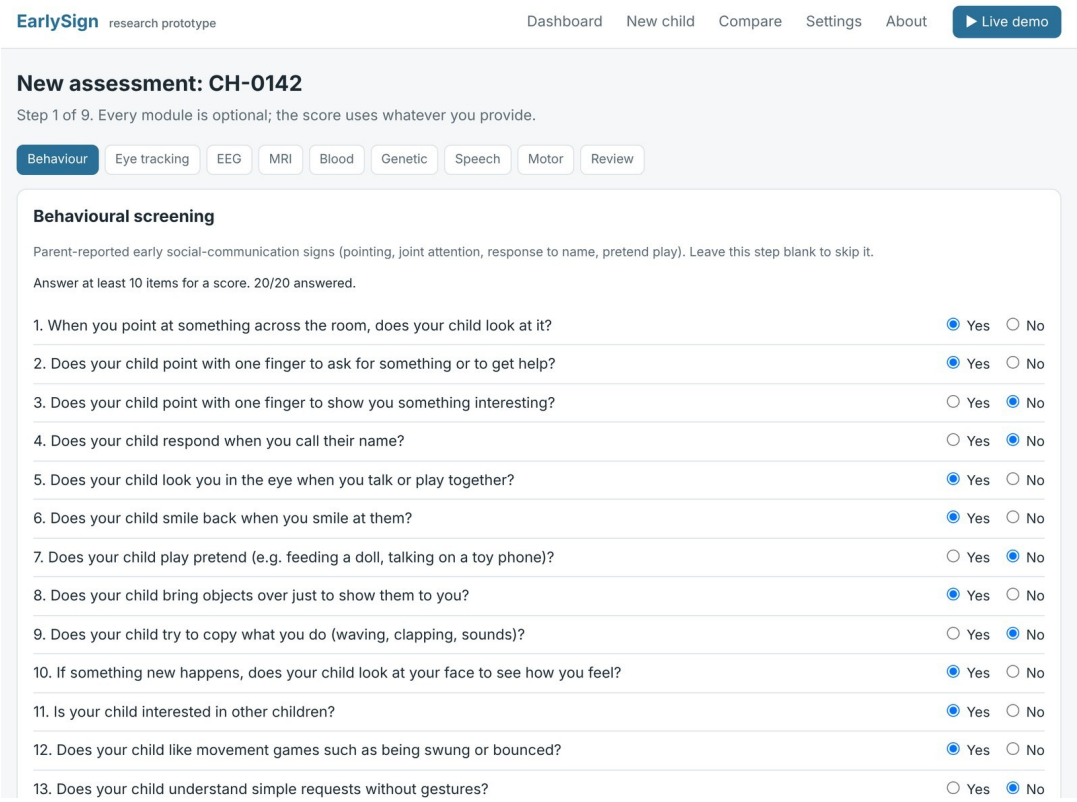


Figure 5 - Behavioural screening questionnaire

Figure 6 shows the eye-tracking step. Each box shows the reference value as a hint. The user can type the numbers or upload a CSV file of gaze data, and the app works out the face and social-scene percentages from the rows.

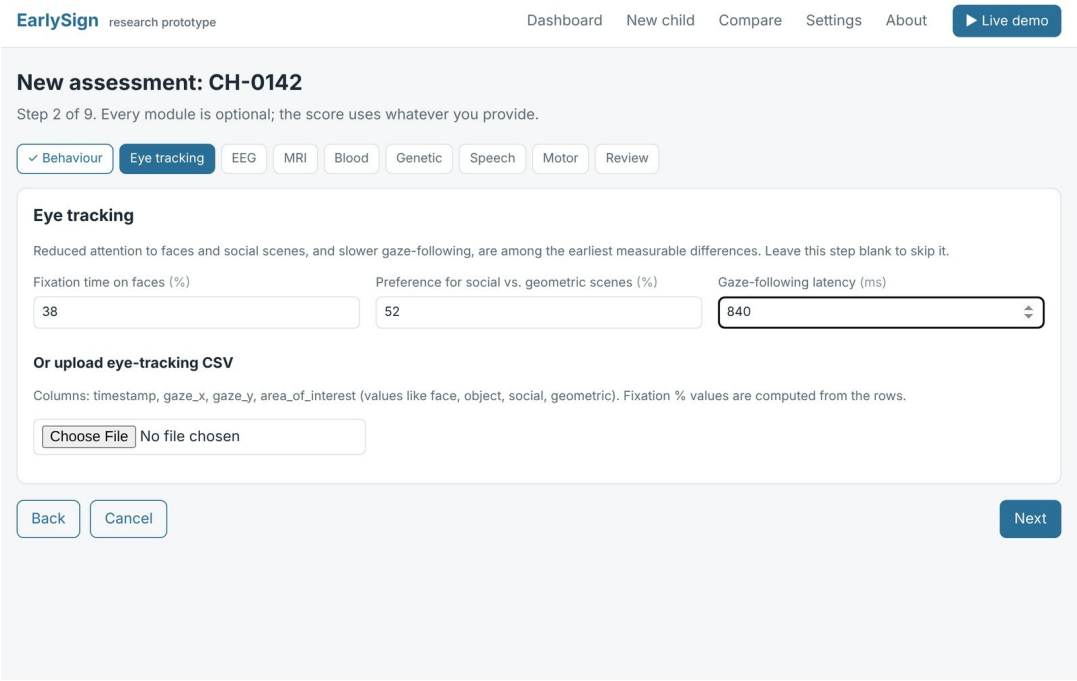


Figure 6 - Eye-tracking step with values entered

The last step, in Figure 7, lists every module with its status and sub-score and shows a preliminary risk level before the user saves. In this example MRI and blood were skipped, and the tool still gives a result from the other six modules.

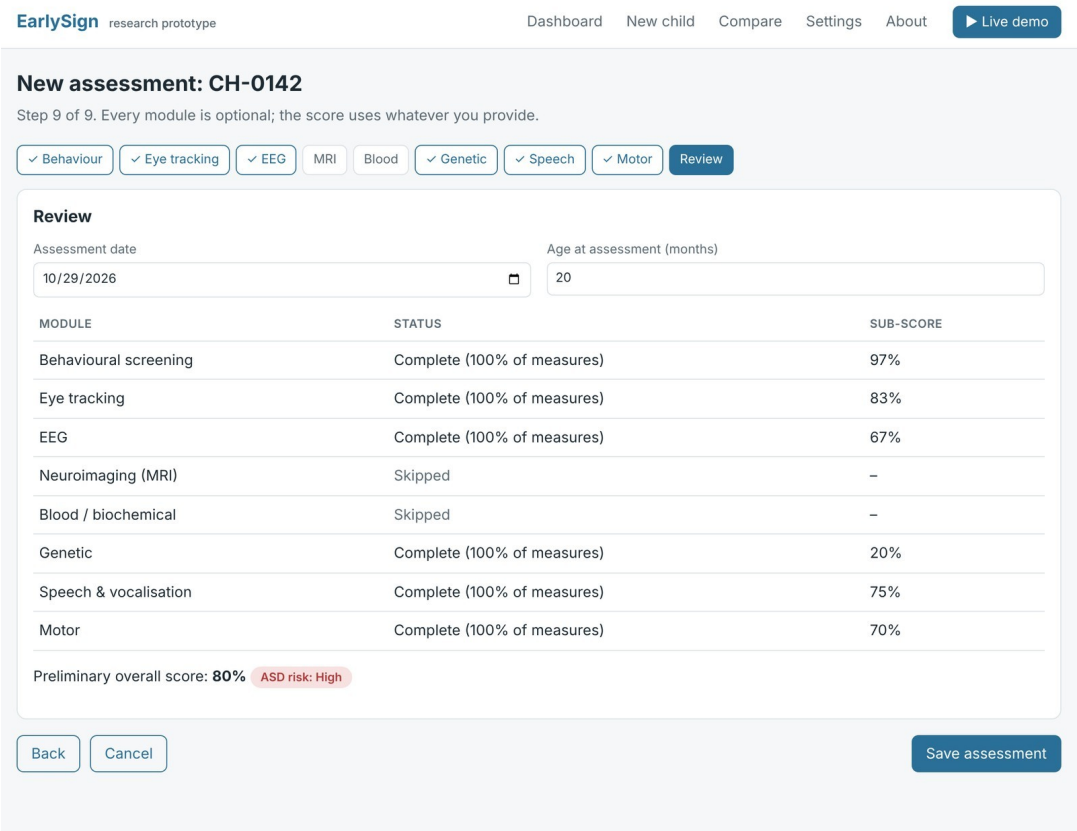


Figure 7 - Review step before saving

4.3 SCREENING REPORT

The report is the main output. At the top it shows the ASD risk level, the score, the data coverage and the suggested next step. Below that, a bar chart shows how much each module added to the score.

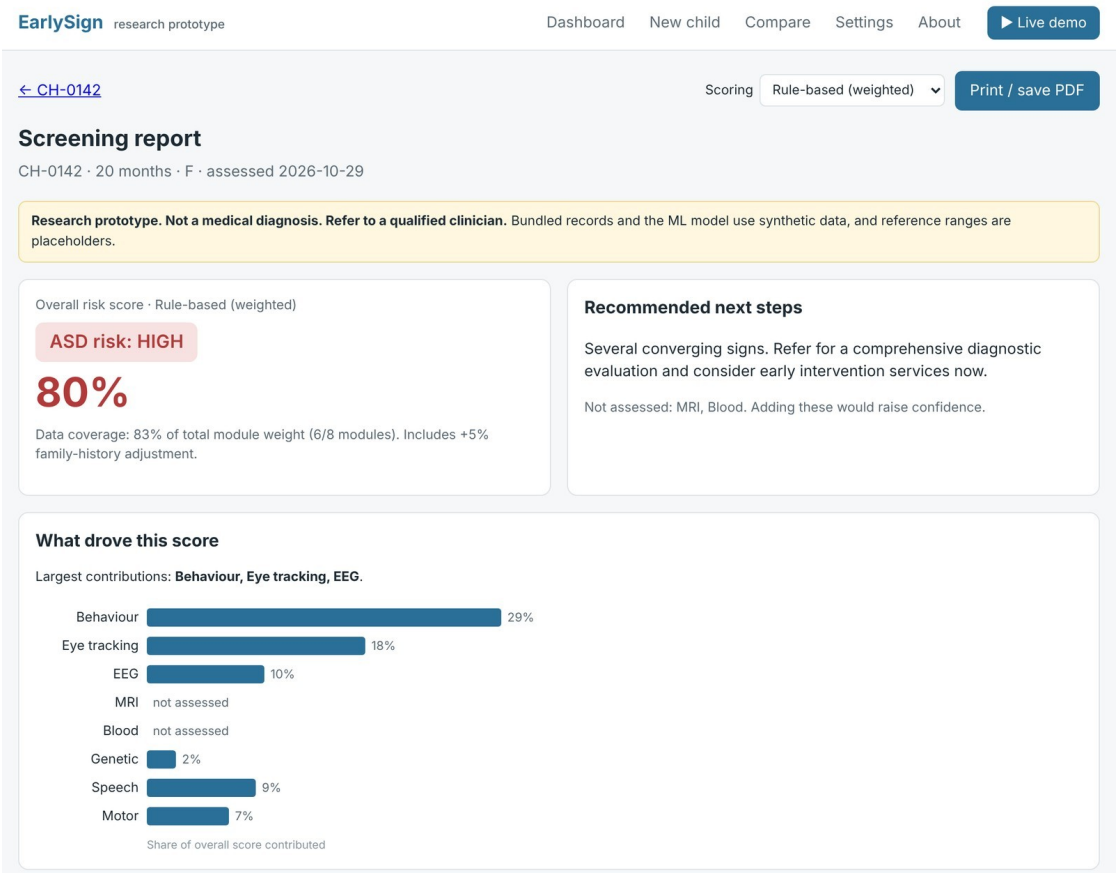
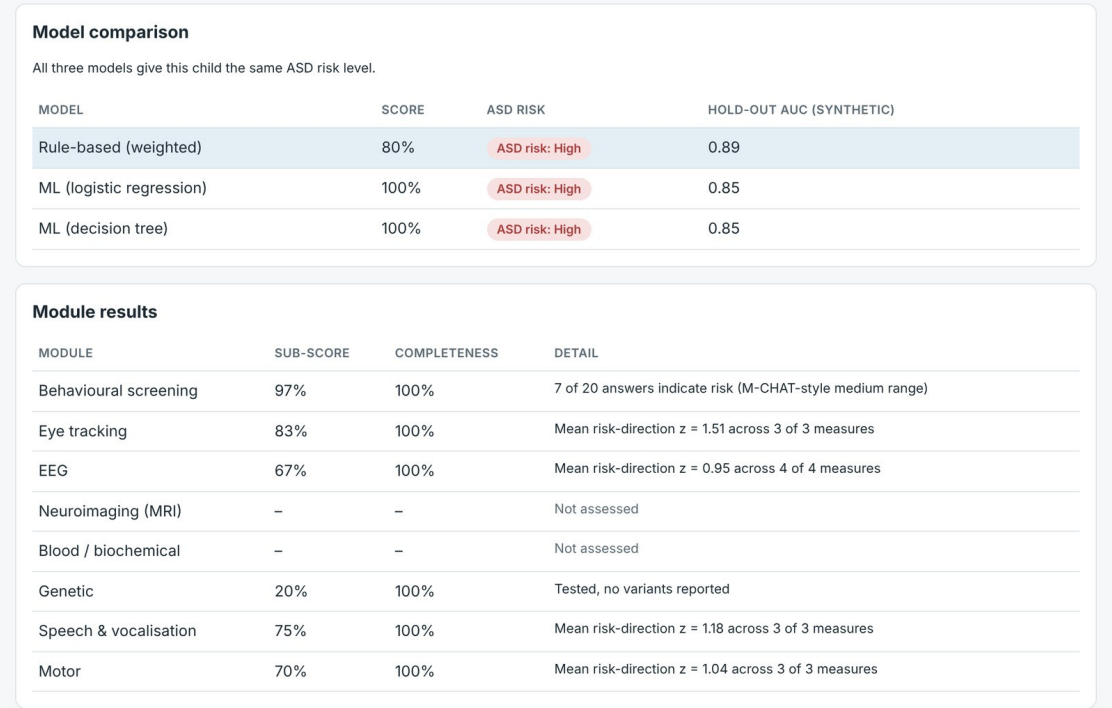


Figure 8 - Screening report with risk level and what drove the score

In Figure 8 the test child CH-0142 is High at 80%. Behaviour, eye tracking and EEG added the most. The coverage line says that 6 of 8 modules were used and that 5 points were added for a sibling with autism. The next-steps box lists the two modules that were not done.

The next part of the report, in Figure 9, shows the score from all three methods side by side. When they disagree, the report says so. It then lists every module with its sub-score, how complete it was and a short detail line.



Model comparison

All three models give this child the same ASD risk level.

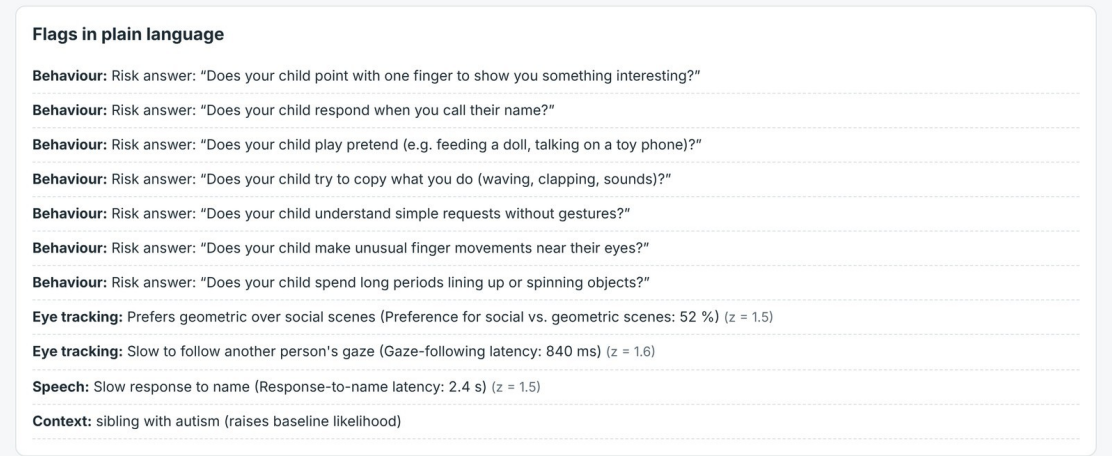
MODEL	SCORE	ASD RISK	HOLD-OUT AUC (SYNTHETIC)
Rule-based (weighted)	80%	ASD risk: High	0.89
ML (logistic regression)	100%	ASD risk: High	0.85
ML (decision tree)	100%	ASD risk: High	0.85

Module results

MODULE	SUB-SCORE	COMPLETENESS	DETAIL
Behavioural screening	97%	100%	7 of 20 answers indicate risk (M-CHAT-style medium range)
Eye tracking	83%	100%	Mean risk-direction z = 1.51 across 3 of 3 measures
EEG	67%	100%	Mean risk-direction z = 0.95 across 4 of 4 measures
Neuroimaging (MRI)	–	–	Not assessed
Blood / biochemical	–	–	Not assessed
Genetic	20%	100%	Tested, no variants reported
Speech & vocalisation	75%	100%	Mean risk-direction z = 1.18 across 3 of 3 measures
Motor	70%	100%	Mean risk-direction z = 1.04 across 3 of 3 measures

Figure 9 - Model comparison and module results

The last part lists the warning signs in plain words, as in Figure 10. Each questionnaire answer that points to risk is shown, along with every measure that was 1.5 standard deviations or more away from its reference.



Flags in plain language

Behaviour: Risk answer: “Does your child point with one finger to show you something interesting?”

Behaviour: Risk answer: “Does your child respond when you call their name?”

Behaviour: Risk answer: “Does your child play pretend (e.g. feeding a doll, talking on a toy phone)?”

Behaviour: Risk answer: “Does your child try to copy what you do (waving, clapping, sounds)?”

Behaviour: Risk answer: “Does your child understand simple requests without gestures?”

Behaviour: Risk answer: “Does your child make unusual finger movements near their eyes?”

Behaviour: Risk answer: “Does your child spend long periods lining up or spinning objects?”

Eye tracking: Prefers geometric over social scenes (Preference for social vs. geometric scenes: 52 %) (z = 1.5)

Eye tracking: Slow to follow another person's gaze (Gaze-following latency: 840 ms) (z = 1.6)

Speech: Slow response to name (Response-to-name latency: 2.4 s) (z = 1.5)

Context: sibling with autism (raises baseline likelihood)

Figure 10 - Warning signs in plain language

When the decision tree is chosen, the chart shows the change in risk at each split instead. Figure 11 shows DEMO-002. The tree started at 29%, which is the share of training children labelled ASD, then followed the path Behaviour > 61%, EEG > 31%, Blood > 24% and ended at 92%. EEG and blood were not done for this child, so they were set to 50% and still took part in the path. We discuss this in Section 4.9.

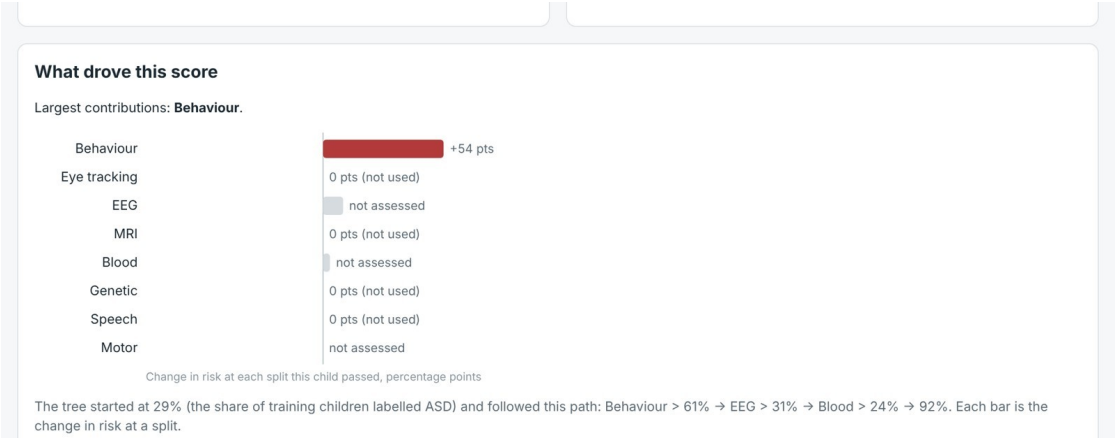


Figure 11 - Decision tree explanation for DEMO-002

4.4 FOLLOW-UP TOOLS

Each child has a page with their family history flags, a chart of risk over time and a list of assessments. The chart is shaded in green, amber and red for the three risk levels. If the score rose by 15 points or more since the last visit, a red trend alert appears at the top.

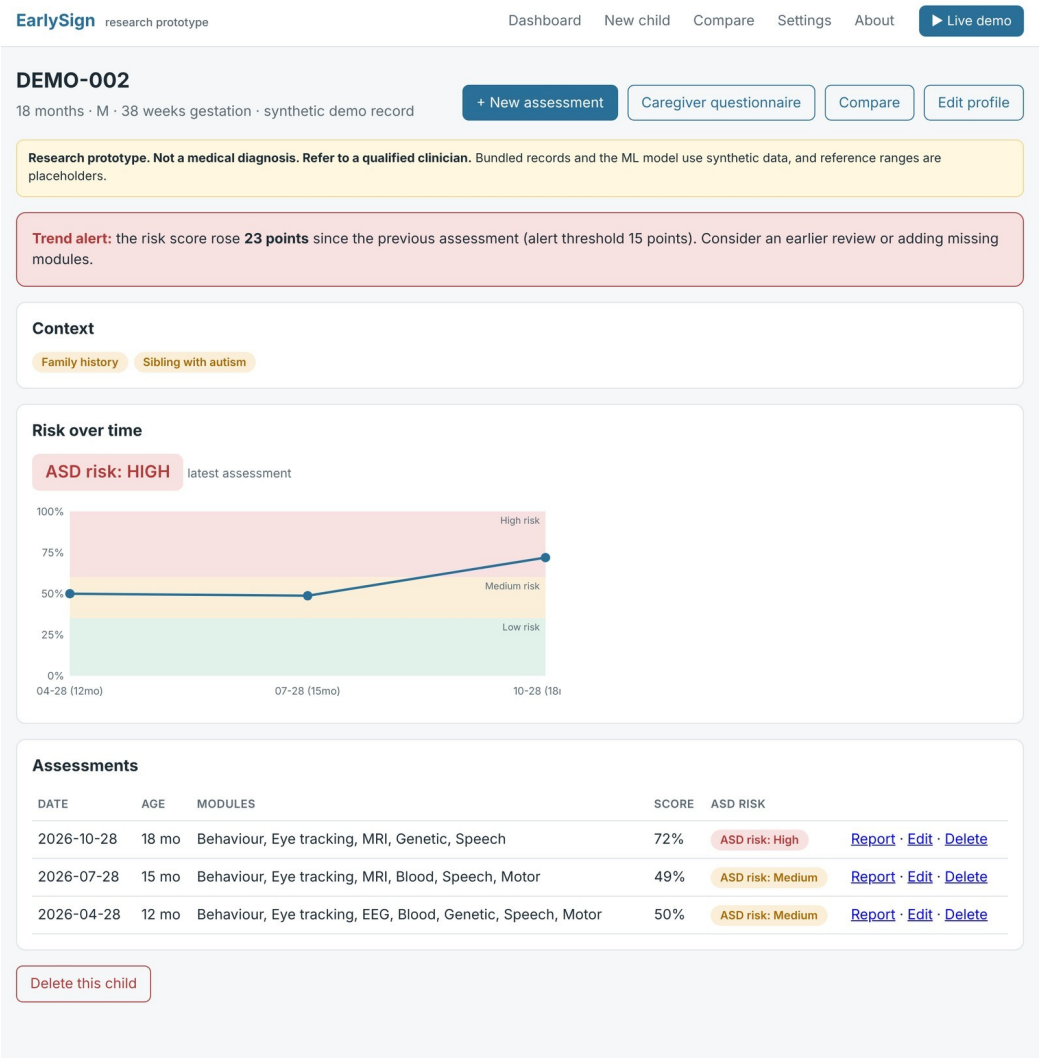


Figure 12 - Child page with risk over time and a trend alert

The compare page in Figure 13 puts two children side by side. It shows their scores under all three methods, their module sub-scores as paired bars and their risk over age on one chart. It also points out the modules that were done for only one of the two children, so the user knows to compare them with care.

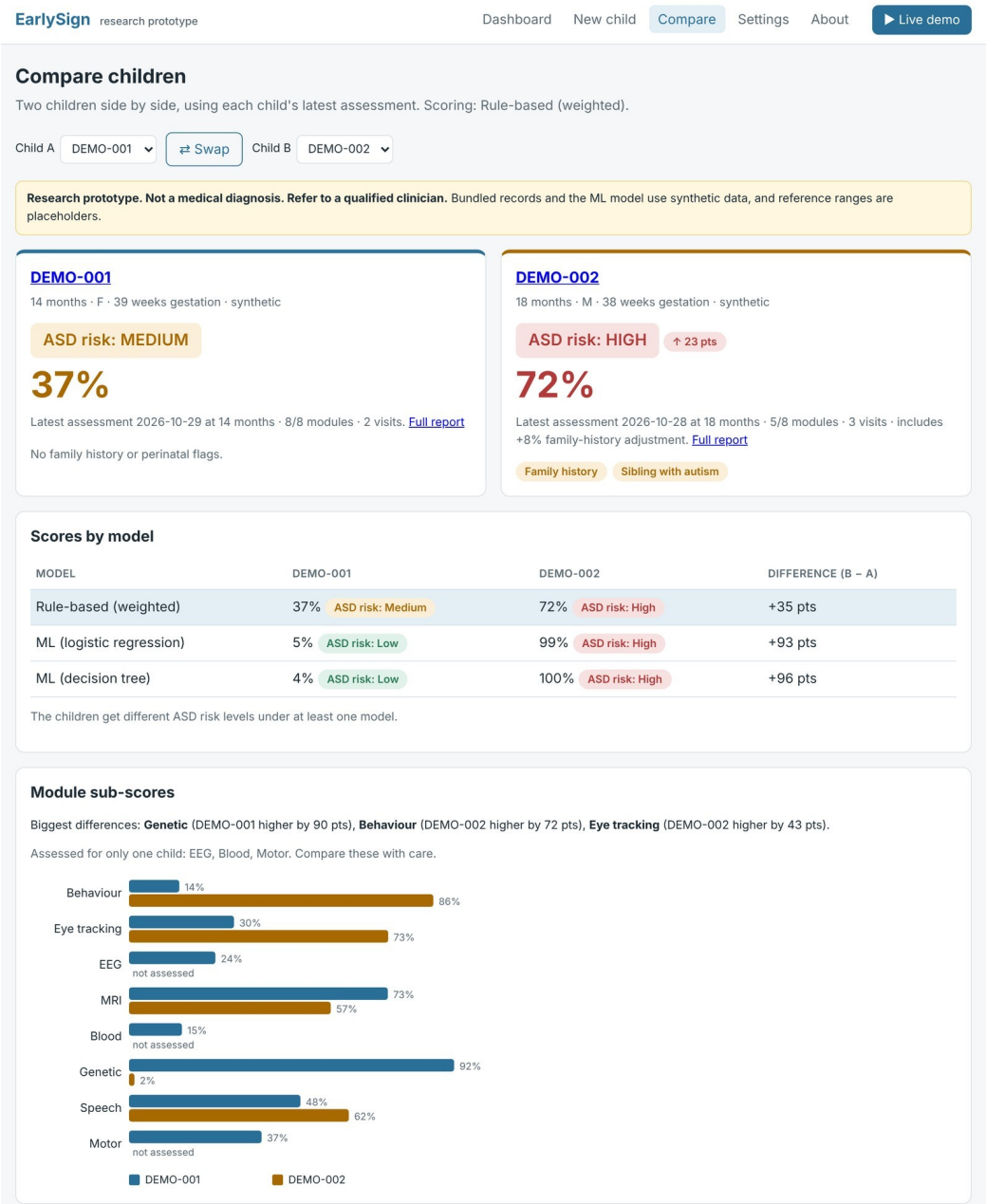


Figure 13 - Comparing two children

Figure 13 also shows that the methods can disagree. DEMO-001 is Medium (37%) on the weighted score but Low on both machine learning models. This is one reason we show all three methods.

The caregiver view in Figure 14 shows only the 20 questions in larger text, without the clinical parts of the app. The doctor can open it on a tablet and hand it to a parent. The answers are saved as a new assessment for that child.

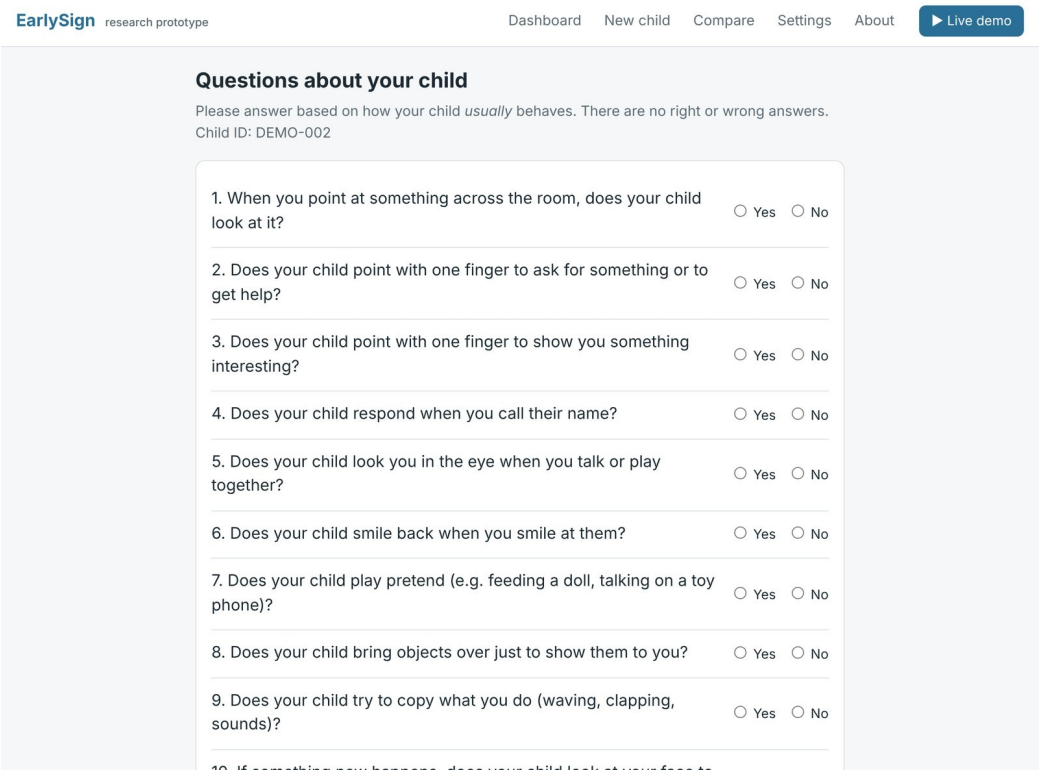


Figure 14 - Caregiver questionnaire view

4.4.1 Step-by-Step Screening of a Child

1. Open EarlySign and click “New child” on the dashboard.
2. Enter a pseudonymous ID, age, sex, gestation and family history, then save.
3. Answer the behaviour questions, or hand the caregiver view to a parent.
4. Enter the results for any other modules that are available. Skip the rest.
5. Check the review step and save the assessment.
6. Read the risk level, the chart of what drove it and the warning signs.
7. Switch between the three scoring methods to see whether they agree.
8. Add a new assessment at the next visit and watch the risk-over-time chart and trend alerts.

4.5 SETTINGS AND MODEL RESULTS

The settings page lets the user choose the scoring method and change the module weights, the risk cut-offs, the family history points, the trend alert limit and every reference range. It also shows what the models learned. Figure 15 shows the logistic regression weights and the six rules found by the decision tree.

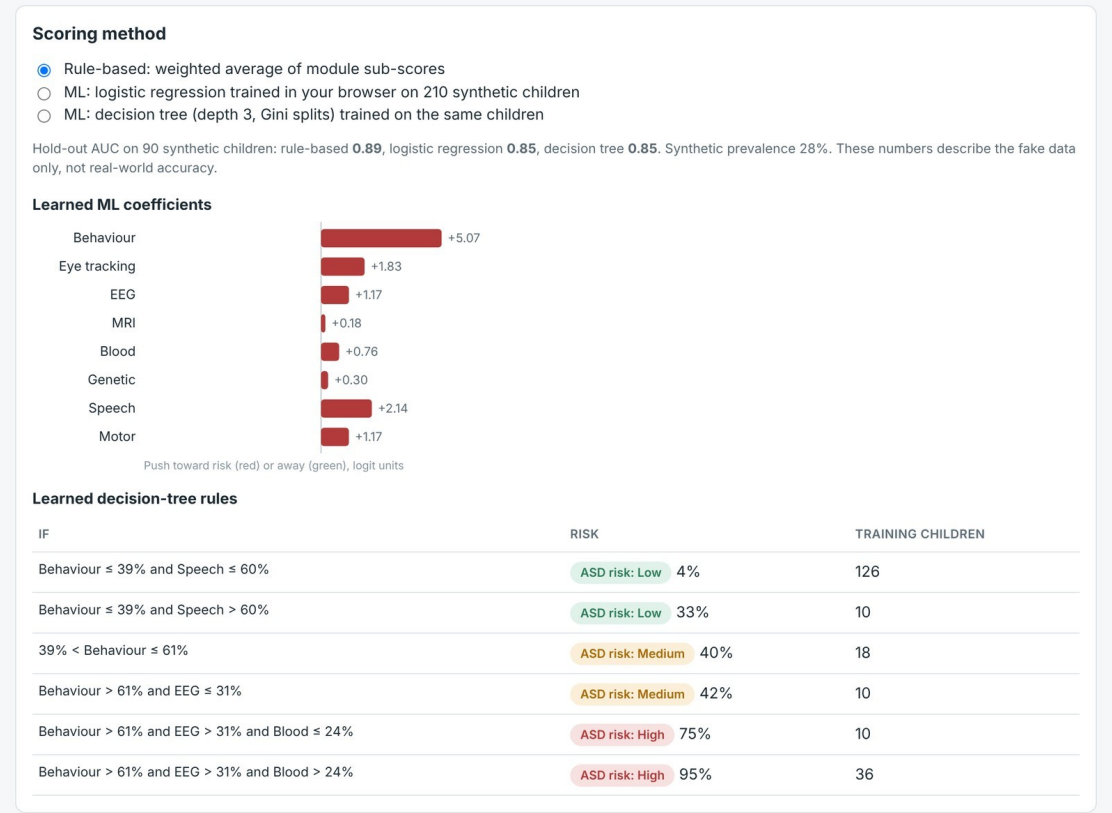
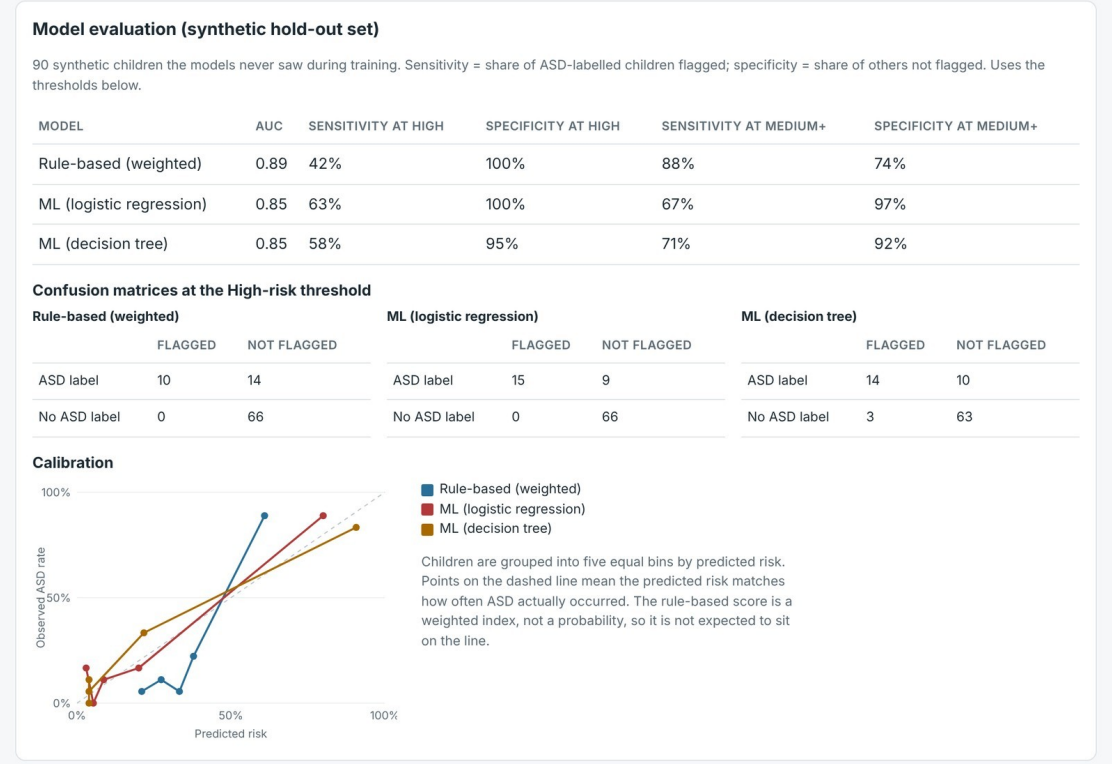


Figure 15 - Learned model weights and decision tree rules

Behaviour has by far the largest weight in the logistic regression (+5.07), followed by speech (+2.14) and eye tracking (+1.83). The tree also splits on behaviour first. This makes sense, because the questionnaire covers 20 answers and so carries the strongest signal in our simulated data.

The same page shows the test results on the 90 held-out children, with confusion matrices and a calibration chart. Figure 16 shows this section. We give the numbers in Section 4.7.



Model evaluation (synthetic hold-out set)

90 synthetic children the models never saw during training. Sensitivity = share of ASD-labelled children flagged; specificity = share of others not flagged. Uses the thresholds below.

MODEL	AUC	SENSITIVITY AT HIGH	SPECIFICITY AT HIGH	SENSITIVITY AT MEDIUM+	SPECIFICITY AT MEDIUM+
Rule-based (weighted)	0.89	42%	100%	88%	74%
ML (logistic regression)	0.85	63%	100%	67%	97%
ML (decision tree)	0.85	58%	95%	71%	92%

Confusion matrices at the High-risk threshold

Rule-based (weighted)

	FLAGGED	NOT FLAGGED
ASD label	10	14
No ASD label	0	66

ML (logistic regression)

	FLAGGED	NOT FLAGGED
ASD label	15	9
No ASD label	0	66

ML (decision tree)

	FLAGGED	NOT FLAGGED
ASD label	14	10
No ASD label	3	63

Calibration

Children are grouped into five equal bins by predicted risk. Points on the dashed line mean the predicted risk matches how often ASD actually occurred. The rule-based score is a weighted index, not a probability, so it is not expected to sit on the line.

Figure 16 - Model evaluation on the synthetic test set

Finally, the “Live demo” button starts a 19-step guided tour, shown in Figure 17. It adds a child, fills in every module, saves the assessment and walks through the report and the compare page. A voice-over can be turned on. Space pauses the tour, the right arrow skips ahead and Esc ends it. We used this for our project reviews.

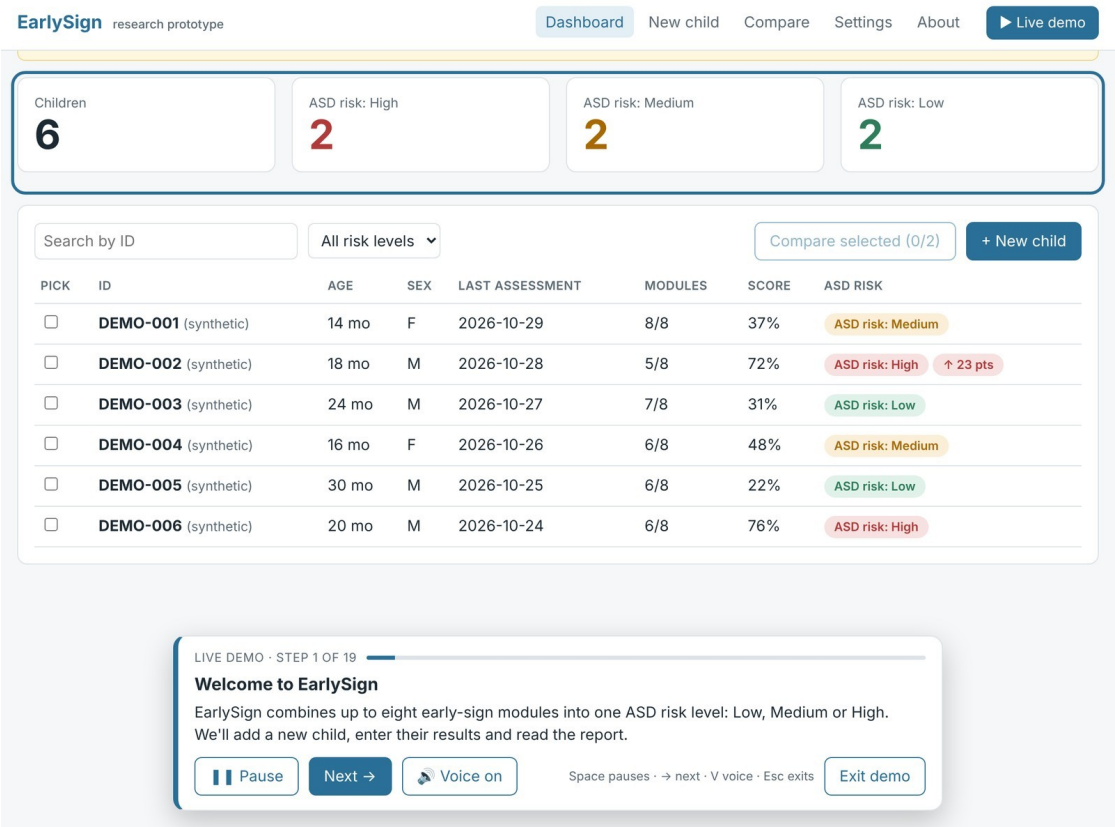


Figure 17 - Live demo with optional voice-over

4.6 CORE LOGIC SUMMARY

The steps below explain how the system works. They are written in plain words and are not copied from the code.

Scoring an assessment

1. For each module that has data, compare each value with its reference mean and standard deviation.
2. Change the sign so that a positive z-score always means more risk.
3. Average the z-scores and pass the result through a logistic curve to get a sub-score.
4. Combine the sub-scores with the chosen method.
5. Add the family history points and turn the score into Low, Medium or High.

Training the models

1. Make 300 simulated children with a fixed seed.
2. Work out the eight sub-scores for each child. Set missing modules to 50%.
3. Train logistic regression and the decision tree on the first 210 children.

4. Test all three methods on the other 90 and work out the AUC, sensitivity and specificity.

Trend alert

1. Sort the child's assessments by date.
2. Score each one with the current method.
3. If the latest score is 15 points or more above the one before, show an alert.

4.7 TESTING AND OBSERVATIONS

We tested the app in two ways. First we ran the main screens in a headless Chromium browser with a script that adds a child, fills every module, saves and opens the report. No errors were raised in the browser console. Table 5 lists what we checked by hand and the figure that shows it.

Table 5 - Test cases and observations

Test case	Figure	Observation
Dashboard counts and trend arrow	Figure 3	Two children in each risk level. DEMO-002 shows a rise of 23 points.
Add a child	Figure 4	The profile saves and the assessment form opens.
Questionnaire scoring	Figure 5	With 20 answers, 7 risk answers give a behaviour sub-score of 97%.
Skipped modules	Figure 7	With MRI and blood skipped, the score uses 6 of 8 modules (83% coverage).
Report and explanation	Figures 8-10	Risk level, contribution chart, model comparison and warning signs all appear.
Tree explanation	Figure 11	The path and the change in risk at each split are shown.
Trend alert	Figure 12	A rise from 49% to 72% raises an alert.
Compare	Figure 13	The methods disagree on DEMO-001 and the page shows it.
Data stays local	-	After a page reload the new child is still there. The file loads no outside scripts and has no code that sends data to a server.

Second, we measured how well the three methods separate the simulated children. The test set has 90 children, of whom 24 have an ASD label. Table 6 gives the results when High (score of 60% or more) is counted as a positive result.

Table 6 - Results on the synthetic test set at the High cut-off

Model	AUC	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-score (%)
Weighted score	0.892	84.4	100.0	41.7	100.0	58.8
Logistic regression	0.851	90.0	100.0	62.5	100.0	76.9
Decision tree	0.852	85.6	82.4	58.3	95.5	68.3

At the High cut-off all three methods are cautious. The weighted score finds only 10 of the 24 children, while logistic regression finds 15 with no false alarms. Missing a child with autism is worse than an extra referral, so we also looked at Medium and above (score of 35% or more). Table 7 gives these results.

Table 7 - Results on the synthetic test set at the Medium cut-off

Model	Sensitivity (%)	Specificity (%)	Precision (%)	Accuracy (%)
Weighted score	87.5	74.2	55.3	77.8
Logistic regression	66.7	97.0	88.9	88.9
Decision tree	70.8	92.4	77.3	86.7

At Medium, the weighted score finds 21 of the 24 children but also flags 17 children without the label. This fits how we meant the levels to be used: Medium means “review again” and High means “refer”. With only 90 test children, small differences between the methods do not mean much. And because the data is simulated by us, these numbers say nothing about how the tool would do with real children.

4.8 SOURCE CODE IMPLEMENTATION

This section shows the core parts of the code as plain text. All of it comes from the one file, index.html. Each part has a short explanation, and Section 4.9 links the code to the results.

Figure 18 - Biomarker module definition

File Name: index.html (eye-tracking module)

```
  { key:"eye", label:"Eye tracking", short:"Eye tracking", csv:true,
    about:"Reduced attention to faces and social scenes, and slower gaze-
following, are among the earliest measurable differences.",
    fields:[
      {key:"faceFix", label:"Fixation time on faces", unit:"%", mean:55, sd:12,
dir:-1, flag:"Less looking at faces than reference"},
      {key:"socialPref", label:"Preference for social vs. geometric scenes",
unit:"%", mean:70, sd:12, dir:-1, flag:"Prefers geometric over social scenes"},
      {key:"gazeLatency", label:"Gaze-following latency", unit:"ms", mean:600,
sd:150, dir:1, flag:"Slow to follow another person's gaze"},
    ]},
```

Explanation: Each module lists its measures. Every measure has a reference mean, a standard deviation, a direction and the warning text shown in the report.

Figure 19 - Module scoring function

File Name: index.html (scoreModule)

```
function scoreModule(mk, data, cfg){
  const m = MOD[mk]; if(!data) return null;
  if(m.kind === "questionnaire"){
    const ans = data.answers || [];
    const answered = ans.filter(a => a==="yes"||a==="no").length;
    if(answered < 10) return null;
    let risk = 0; const flags = [];
    QUESTIONS.forEach((q,i) => { if(ans[i] === q[1]){ risk++;
flags.push({text:"Risk answer: “"+q[0]+"”", z:null}); } });
    const scaled = risk * QUESTIONS.length / answered;
    const band = scaled <= 2 ? "low" : scaled <= 7 ? "medium" : "high";
    return { score:sigmoid(0.9*(scaled-3)), conf:answered/QUESTIONS.length, flags,
detail:`${risk} of ${answered} answers indicate risk (M-CHAT-style ${band} range)`
};
  }
  const zs = []; const flags = []; let filled = 0;
  for(const f of m.fields){
    const v = data[f.key]; if(!has(v)) continue;
    filled++;
    const r = cfgRange(cfg, mk, f);
    const z = (Number(v) - r.mean) / r.sd * f.dir;
    zs.push(z);
    if(z >= 1.5) flags.push({text:`${f.flag} (${f.label}: ${v}${f.unit&&!
f.unit.includes("–")?" "+f.unit:""})`, z});
  }
  if(m.kind === "genetic"){
    const vars = (data.variants||[]);
    if(!data.tested && !vars.length && !zs.length) return null;
    let s = zs.length ? sigmoid(1.6*(zs[0]-0.5)) : 0.2;
    if(vars.length){ s = Math.max(0.85, s); vars.forEach(v =>
flags.unshift({text:`Known high-impact variant reported: ${v}`, z:null})); }
    return { score:s, conf:1, flags, detail: vars.length ? `${vars.length}
variant(s) reported` : (zs.length ? `Polygenic z = ${fmt(zs[0])}` : "Tested, no
variants reported") };
  }
  if(!filled) return null;
  const avg = zs.reduce((a,b)=>a+b,0)/zs.length;
  return { score:sigmoid(1.6*(avg-0.5)), conf:filled/m.fields.length, flags,
detail:`Mean risk-direction z = ${fmt(avg)} across ${filled} of ${m.fields.length}
measures` };
}
```

Explanation: This function turns one module's raw values into a sub-score. The questionnaire, the genetic module and the other modules each have their own rule.

Figure 20 - Logistic regression training in the browser

File Name: index.html (trainLR)

```
function trainLR(X, y, {epochs=2000, lr=1.0, l2=0.002}={}){
  const d = X[0].length; let w = new Array(d).fill(0), b = 0;
  for(let e=0;e<epochs;e++){
    const gw = new Array(d).fill(0); let gb = 0;
    for(let i=0;i<X.length;i++){
      const p = sigmoid(b + X[i].reduce((s,x,j)=>s+x*w[j],0)); const err = p -
y[i];
      for(let j=0;j<d;j++) gw[j] += err*X[i][j]; gb += err;
    }
    for(let j=0;j<d;j++) w[j] -= lr*(gw[j]/X.length + l2*w[j]);
    b -= lr*gb/X.length;
  }
  return {w, b};
}
```

Explanation: Plain gradient descent with a small L2 penalty. It runs 2,000 rounds over the 210 training children when the page loads.

Figure 21 - Decision tree explanation

File Name: index.html (explainTree)

```
// Walk one child down the tree; each split's change in risk is credited to the
biomarker it tested (Saabas attribution).
function explainTree(tree, x){
  const contrib = new Array(x.length).fill(0), path = [];
  let nd = tree;
  while(nd.left){
    const goLeft = x[nd.feat] <= nd.thr, next = goLeft ? nd.left : nd.right;
    contrib[nd.feat] += next.value - nd.value;
    path.push({feat:nd.feat, thr:nd.thr, le:goLeft});
    nd = next;
  }
  return { value:nd.value, contrib, path };
}
```

Explanation: The child walks down the tree. At each split, the change in risk is added to the module that the split tested. This gives the bars in Figure 11.

Figure 22 - Eye-tracking CSV reader and trend alert

File Name: index.html (parseEyeCSV, trendAlert)

```
function parseEyeCSV(text){
  const lines = text.trim().split(/\r?\n/).filter(Boolean);
  if(lines.length < 2) return {error:"CSV has no data rows."};
  const head = lines[0].split(",").map(h => h.trim().toLowerCase());
  const ai = head.indexOf("area_of_interest");
  if(ai < 0) return {error:"Missing an area_of_interest column."};
  const aois = lines.slice(1).map(l => (l.split(",")
[ai]||"").trim().toLowerCase());
  const cnt = k => aois.filter(a => a === k).length;
  const values = {};
  if(cnt("face")+cnt("object")) values.faceFix = +(100*cnt("face")/(cnt("face")
+cnt("object"))).toFixed(1);
  if(cnt("social")+cnt("geometric")) values.socialPref =
+(100*cnt("social")/(cnt("social")+cnt("geometric"))).toFixed(1);
  if(!Object.keys(values).length) return {error:"No face/object or
social/geometric rows found."};
  return {values, n:aois.length};
}


// Rise in score between the last two assessments, if above the alert threshold.
function trendAlert(c, cfg){
  const list = c.assessments.slice().sort((a,b)=>a.date<b.date?-1:1).map(a =>
scoreAssessment(c, a, cfg, MODEL).overall).filter(v => v!=null);
  if(list.length < 2) return null;
  const rise = list[list.length-1] - list[list.length-2];
  return rise >= (+cfg.trendAlert || 0.15) ? rise : null;
}
```

Explanation: The CSV reader counts rows by area of interest to get the face and social-scene percentages. The trend alert compares the last two scores of a child.

4.9 RESULTS AND DISCUSSION

The prototype does what we set out to do. A user can enter any mix of test results, get one risk level, and see why the tool reached it. The three methods are shown side by side, and the compare page and trend alert help with follow-up.

Our results suggest that a simple weighted score can be as useful as machine learning when the data is small. The weighted score had the best AUC (0.89). Logistic regression had the best balance at the High cut-off. The decision tree is the easiest to read, because its six rules fit on one screen. We would not pick a winner from 90 simulated children.

We also found a weakness while testing. Both models set a missing module to 50%. For the decision tree this can matter a lot. In Figure 11, EEG and blood were not done for DEMO-002, yet the tree used them in its path because 50% happened to be above the split values. A better approach would be to train the models with missing values in mind, or to stop the tree at a split on a missing module.

Another point is that the weighted score is not a probability. The calibration chart in Figure 16 shows that a weighted score of 60% does not mean a 60% chance of

autism. The logistic regression is closer to a probability. For this reason we call the output a “risk level” and not a chance.

Table 8 links each result to the code that produced it.

Table 8 - Results and the code behind them

Result	Figure	Main code function(s)
Dashboard and trend arrow	Figure 3	viewDashboard, trendAlert
Child profile and assessment form	Figures 4-7	viewNewChild, viewAssess, parseEyeCSV
Risk level and explanation	Figures 8-10	scoreModule, scoreAssessment, viewResult
Decision tree path	Figure 11	trainTree, explainTree
Risk over time and compare	Figures 12-13	viewChild, viewCompare, lineChart
Model weights, rules and test results	Figures 15-16	trainLR, treeRules, auc, confusion, calibration
Live demo	Figure 17	tourPlay, tourSpeak

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

We built EarlySign, a working screening prototype that combines eight kinds of early signs of autism into one risk level of Low, Medium or High. It works with whatever tests are available, explains every result and shows a simple weighted score next to two machine learning models. It also tracks children over repeat visits, compares two children and raises an alert when risk rises sharply.

On 90 simulated test children, the three methods reached an AUC between 0.85 and 0.89. These results show that the design works from end to end. They do not show that it works on real children, because the data was made by us.

Our main aim is to help doctors decide sooner which toddlers should get a full assessment. EarlySign cannot diagnose autism and should not replace a clinical assessment.

5.2 LIMITATIONS

- All training and test data is simulated. The tool has not been tried with real patients.
- The reference ranges and weights are our own placeholders, not clinical norms for each age and sex.
- The test set has only 90 children, so small differences between methods are not reliable.
- Missing modules are set to 50% in the models, which can push the decision tree the wrong way.
- The tool takes summary numbers and cannot read raw EEG, MRI or audio files.
- Several modules need costly lab tests that are not routine for toddlers.

5.3 FUTURE ENHANCEMENTS

- Test the tool on real, consented clinical data with approval from an ethics committee.
- Replace the placeholder ranges with proper norms for each age and sex.
- Work out features directly from raw eye-tracking, EEG and voice recordings.
- Handle missing modules properly in the models instead of setting them to 50%.

- Try stronger models such as gradient boosting, with calibration so the output is a real probability.
- Run a small pilot with a children's clinic and add the questionnaire in Tamil and other Indian languages.

5.4 FINAL REMARK

EarlySign shows that early signs of autism from very different sources can be brought together on one screen and explained in plain words. The next step is to test it with real children, so that it can one day help families get support sooner.

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